

Robust Inference on Macro Equations with Shock Proxies

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Structural macroeconomic equations and endogeneity

- Consider a linear(ized) structural macro equation represented as

$$y_t = w_t' \delta + u_t, \quad E[w_t u_t] \neq 0$$

- Take the New Keynesian Phillips curve as an example:

$$\pi_t = \lambda x_t + \gamma_f E_t(\pi_{t+1}) + \varepsilon_t^c$$

Two sources of endogeneity

- Measurement error in x_t and $E_t(\pi_{t+1})$
 - Simultaneous-equation bias due to ε_t^c
- Instrumental variables...? Predetermined variables may not be exogenous (c.f., DSGE) (Hansen and Singleton 1982, Galí and Gertler 1999)

A new approach in the literature: shock proxies as IV

- Barnichon and Mesters (2020 QJE): A sequence of **shock proxies**
 - Exogeneity: Proxies are more likely to satisfy exogeneity
 - Relevance: Weak instruments. **Anderson-Rubin (AR) test**
 - Many instruments: Almon parameterization
- Still, the econometric toolkit is far from complete
 - ① The AR statistic has some disadvantages under **overidentification** (DF, rejection against overidentifying restrictions)
 - ② Barnichon and Mesters (2020) tend to **over-reject the null**

My proposal: KLM and the restricted long-run variance

- 1 The **Almon-parametrized KLM** (Kleibergen) statistic:
 $DF = \# \text{ str params.}$ Pivotal under the weak IV asymptotics
- 2 Apply the **restricted HAR long-run variance**
Accurate size control for both the AR and KLM statistics

Two applications: the Phillips curve and fiscal rules

- Does the size-accuracy matter in practice? Yes, it does!
- With the proposed statistic, either a completely flat or a very steep Phillips curve is no longer rejected (cf. Barnichon and Mesters 2020)
- Finite-sample size distortion can result in an opposite conclusion
- I also study fiscal policy rules and multipliers, and find a statistically significant tax multiplier

Related Literature

- Weak-instrument-robust inference on macro equations:
Barnichon Mesters (2020), Lewis Mertens (2024); Euler Eq: Yogo (2004), Ascari Magnusson Mavroeidis (2021); NKPC: Kleibergen Mavroeidis (2009), Magnusson Mavroeidis (2014), Mavroeidis Plagborg-Møller Stock (2014); Taylor rule: Mavroeidis (2010), Carlevaro Haque Magnusson (2025); among many many others
Overidentification. Size accuracy. Widely applicable method
- HAR inference and finite-sample size distortion:
Lazarus Lewis Stock Watson (2018), Vogelsang (2018); Newey West (1986), Andrews (1991); Keifer Vogelsang (2000, 2002, 2005), Jansson (2004), Sun (2014)...
Weak-IV robust inference. Restricted variance. Fixed-b
- Fiscal rule/multiplier estimation:
Caldara Kamps (2017), Mertens Ravn (2013, 2014), Angelini Caggiano Castelnovo Fanelli (2023), Keweloh Klein Prüser (2025); Montiel Olea Mertens (2018), Hebous and Zimmermann (2018)...
Complements proxy-SVAR approaches. First statistically significant evidence on the tax multiplier with weak instruments

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- 5 Empirical application: Fiscal policy rules and multipliers
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The model in one slide

- Consider the linear model

$$y_t = w_t' \delta + u_t$$

where $E[w_t u_t] \neq 0$, $\dim(\delta) = m \leq 3$

- Suppose there exists a sequence of shock proxies

$$\xi_{t:t-H} = (\xi_t, \dots, \xi_{t-H})'$$

where $E[u_t \xi_s] = 0, \forall t, s$ (relevance condition discussed later)

- Apply the Almon parametrization

$$z_t = \left(\sum_{h=0}^H \xi_{t-h}, \sum_{h=0}^H h \xi_{t-h}, \sum_{h=0}^H h^2 \xi_{t-h} \right)'$$

and assume

$$w_t = \Pi' z_t + v_t$$

How to choose H ? Why the Almon parametrization?

“Regression in impulse response space”

- Regression in impulse response space (Barnichon and Mesters, 2020)

$$\mathcal{R}_h^\pi = \gamma_f \mathcal{R}_{h+1}^\pi + \lambda \mathcal{R}_h^x, \text{ for } h = 0, 1, 2, \dots, H.$$

where $\mathcal{R}_h^y = E(y_t \xi_{t:t-H})$ is the vector of impulse responses of variable y to the shock ξ

- Empirical IRFs are typically different from zero for three to five years: $H \in [12, 20]$ for quarterly data provides information for identification but could be “too many”
- The Almon-param mitigates the many-instrument problem
- Smooth macro IRFs. A shape restriction: the IRF is quadratic
$$u_t = \theta^{(0)} \xi_t + \theta^{(1)} \xi_{t-1} + \dots \theta^{(H)} \xi_{t-H}, \quad \theta^{(h)} = a + bh + c^2 h$$
$$\Leftrightarrow u_t = \left(\sum_{h=0}^H \xi_{t-h}, \sum_{h=0}^H h \xi_{t-h}, \sum_{h=0}^H h^2 \xi_{t-h} \right) \theta_a, \theta_a = (a, b, c)'$$

Weak IV robust inference with Almon parametrization

- Goal: Construct confidence sets robust to weak instruments
- How: Test inversion
- Weak-instrument robust test: correct asymptotic coverage for $H_0 : \delta = \delta_0$ regardless of the instrument strength
- ① Just-identification: the Almon-parameterized AR statistic
Barnichon and Mesters (2020)
- ② Over-identification: the Almon-parameterized **KLM** statistic,
 $DF = m < k = 3$
- Applying **the restricted HAR long-run variance** is important for size control
- Next: Test statistics, asymptotics, and finite-sample properties

AR test with the restricted long-run variance

- Intuition: Consider regression $u_t(\delta_0) := y_t - w_t'\delta_0 = z_t'\theta_a + \eta_t$. θ_a should be zero if the exogeneity condition holds

$$AR_a^r(\delta_0) = \hat{\theta}_a' \tilde{\Sigma}_{\theta_a}^{-1} \hat{\theta}_a = \frac{(y - W\delta_0)' P_Z (y - W\delta_0)}{\tilde{s}_u^2},$$

where

$$\hat{\theta}_a = (Z'Z)^{-1} Z'(y - W\delta_0)$$

$$\tilde{\Sigma}_{\theta_a} = (Z'Z)^{-1} \tilde{s}_u^2$$

$$\tilde{s}_u^2 = \frac{1}{n-H} \sum_{t=H+1}^n \sum_{s=H+1}^n \tilde{u}_t \tilde{u}_s \kappa((t-s)/b_n), \quad \kappa(\cdot) \text{ is some kernel,}$$

$$\tilde{u}_t = \tilde{\tilde{u}}_t - \frac{1}{n-H} \sum_{t=H+1}^n \tilde{\tilde{u}}_t, \quad \tilde{\tilde{u}}_t = (y_t - w_t'\delta_0) - z_t'\mathbf{0}.$$

AR test with the unrestricted long-run variance

- Barnichon and Mesters (2020) use the **unrestricted** long-run variance, $\hat{s}_u^2$: AR_a
- Just replace $\tilde{u}_t$ with the following unrestricted version

$$\hat{u}_t = (y_t - w_t' \delta_0) - z_t' \hat{\theta}_a$$

The limiting distribution of the AR test statistics

Theorem (Barnichon and Mestes (2020))

Let [Assumption 1](#) hold. Under $H_0: \delta = \delta_0$ for $\frac{H}{n} \rightarrow c \in (0, 1)$ as $n \rightarrow \infty$,

$$AR_a(\delta_0) \Rightarrow \chi^2(3).$$

No restrictions on Π in Assumption 1: Weak-IV robust

Corollary

Under the same conditions above,

$$AR_a^r(\delta_0) \Rightarrow \chi^2(3).$$

KLM statistic: overidentified models

- The Almon-parametrized KLM statistic:

$$K_a^r(\delta_0) = \frac{(y - W\delta_0)' P_{Z\tilde{\Pi}(\delta_0)} (y - W\delta_0)}{\tilde{s}_u^2}$$

where

$$\tilde{\Pi}(\delta_0) = (Z'Z)^{-1}Z' \left[W - (y - W\delta_0) \frac{\tilde{s}_u V}{\tilde{s}_u^2} \right]$$

- Intuition: Projection on $Z\tilde{\Pi}$, instead of Z ; Decorrelation trick.
Quadratic form of the score of the concentrated log-likelihood
- $K_a(\delta_0)$ is defined similarly by using the unrestricted variance $\hat{s}_u^2$

The limiting distribution of the KLM statistics

- Three types of instrument validity:
 - (i) Instruments are relevant: Π is fixed and full rank
 - (ii) Instruments are weakly relevant: $\Pi = \Pi_n = K_n^{-1}C$ with C fixed and full rank, and $K_n = \text{diag}(n, n^2, n^3)$
 - (iii) Instruments are irrelevant: $\Pi = 0$

Theorem

Let Assumption 1 hold. Under $H_0: \delta = \delta_0$ for $\frac{H}{n} \rightarrow c \in (0, 1)$ as $n \rightarrow \infty$,

$$K_a^r(\delta_0), K_a(\delta_0) \Rightarrow \chi^2(\textcolor{red}{m})$$

for all (i), (ii), (iii) cases.

Case (ii) is equivalent to the Staiger-Stock framework

- Case (ii) Instruments are weakly relevant: $\Pi = \Pi_n = K_n^{-1}C$ with C fixed and full rank, and $K_n = \text{diag}(n, n^2, n^3)$
- This is not “super” weak, but the standard local to zero weak-instrument asymptotics with Almon parameterization
- The signal component of the concentration parameter does not diverge: $\mu := \Pi'(Z'Z)\Pi = C'(K_n^{-1}Z'ZK_n^{-1})C \Rightarrow C'\Psi C$
- The signal of additional information is the same

$$\pi_n \xi_t = cn^{-1/2} \underbrace{\xi_t}_{O_p(1)} = O_p(n^{-1/2})$$

$$\Pi_n' z_t = C' \begin{bmatrix} n^{-1/2} & 0 & 0 \\ 0 & n^{-1/2} & 0 \\ 0 & 0 & n^{-1/2} \end{bmatrix} \underbrace{\begin{bmatrix} n^{-1/2} \sum_{h=0}^H \xi_{t-h} \\ n^{-3/2} \sum_{h=0}^H h \xi_{t-h} \\ n^{-5/2} \sum_{h=0}^H h^2 \xi_{t-h} \end{bmatrix}}_{O_p(1) \text{ by FCLT}} = O_p(n^{-1/2})$$

Finite-sample properties

- $\{AR_a^r, AR_a\}$ and $\{K_a^r, K_a\}$ are asymptotically equivalent, but have **different finite-sample properties**
- The HAR inference literature: standard inference—using the **unrestricted** long-run variance with χ^2 **critical values**—induces over-rejection in finite samples (Lazarus, Lewis, Stock, Watson 2018). This applies to AR_a of Barnichon and Mesters (2020)
- Using the **restricted** long-run variance helps to reduce finite-sample size distortion (LLSW 2018; Vogelsang 2018). This applies to AR_a^r and K_a^r
- Combining AR_a and K_a with the fixed- b asymptotics may also help to control size (a series of Kiefer and Vogelsang; Sun 2014)

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Simulation setup

- DGP based on a hybrid New Keynesian Phillips curve:

$$\pi_t = \gamma_b \pi_{t-1} + \gamma_f E_t(\pi_{t+1}) + \lambda x_t + e_t$$

$$x_t = \rho_1 x_{t-1} + \rho_2 x_{t-2} + \varepsilon_t + \nu e_t$$

where $e_t = \rho e_{t-1} + \sqrt{1 - \rho^2} \zeta_t$ with $\zeta_t \sim N(0, 1)$ and $\varepsilon_t \sim N(0, \sigma_i^2)$

- $[\gamma_b, \gamma_f] = [0.6, 0.4]$, implying $\gamma_b + \gamma_f = 1$, a common restriction in empirical studies (also $[\gamma_b, \gamma_f] = [0.6, 0.3]$ in the paper)
- Transform the model by imposing the restriction: $\delta = (\gamma_f, \lambda)$
- Run 10,000 simulations and compute the rejection frequency with different ρ and σ_i

Simulation results with $\gamma_b + \gamma_f = 1$ ($\alpha = 5\%$, $T = 100$)

σ_i	ρ	AR_a^r	AR_a	K_a^r	K_a
0.10	0.0	5.0	10.4	3.5	8.8
0.25	0.0	4.9	10.4	3.9	8.7
0.50	0.0	5.4	10.1	4.8	8.8
1.00	0.0	6.6	9.5	5.7	8.2
0.10	0.5	5.5	12.4	3.5	10.0
0.25	0.5	5.1	12.1	3.3	9.5
0.50	0.5	5.3	12.1	3.5	9.8
1.00	0.5	6.3	11.6	4.4	9.6

- AR_a^r and K_a^r have accurate size, while AR_a and K_a over-reject
- Robust when $\alpha = 0.1, 0.32$ and without $\gamma_b + \gamma_f = 1$ for AR_a, AR_a^r
- Both K^r and K overreject: The Almon-parametrization is necessary
- Fixed- b improves AR_a, K_a , but too conservative for AR_a^r, K_a^r

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Phillips curve

- Analyze the hybrid New Keynesian Phillips curve

$$\pi_t = \gamma_b \pi_{t-1} + \gamma_f E_t(\pi_{t+1}) + \lambda x_t + \varepsilon_t^s$$

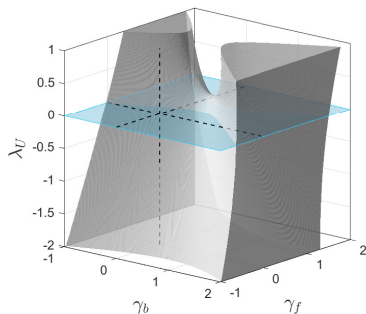
with and without the common restriction $\gamma_b + \gamma_f = 1$

- Instruments are monetary policy shock proxies

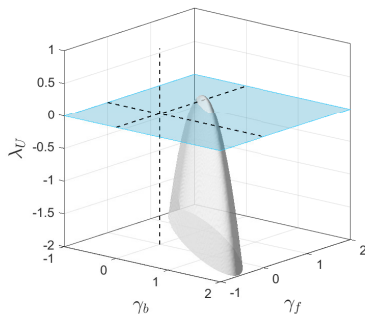
Data

- I use the same dataset as Barnichon and Mesters (2020)
- Variables:
 - Inflation: PCE excluding food and energy prices
 - Unemployment gap: HP-detrended unemployment rate
 - Output gap: HP-detrended GDP
- Instruments:
 - 1969–2007 Romer–Romer narrative series
 - 1990–2017 High-frequency identified (HFI) series:
 Δ_{FOMC} 3-month-ahead FF futures + Δ_{FOMC} 10-year yield
- The weak-IV test of Lewis and Mertens (2025): these proxies are not sufficiently strong [Results](#)

1969–2007: The Romer-Romer shock



(a) AR_a^r with $\chi^2(3)$

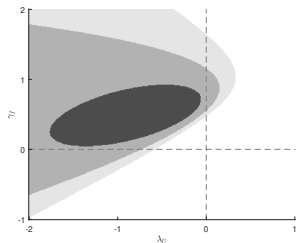


(b) AR_a with $\chi^2(3)$

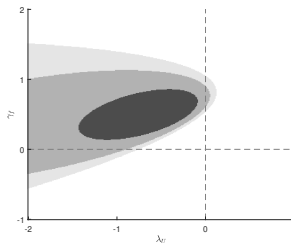
Figure: 95% confidence sets

- Much wider confidence sets with AR_a than AR_a^r
- May reach an opposite conclusion between AR_a^r and AR_a regarding the significance of λ_U and the area of γ_f and γ_b

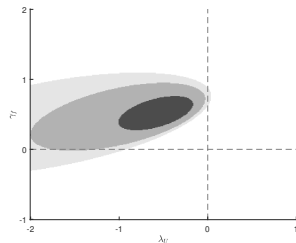
Similar results with the common restriction $\gamma_b + \gamma_f = 1$



(a) AR_a^r with $\chi^2(3)$



(b) AR_a with $\chi^2(3)$



(c) Barnichon and Mesters

- Non-significant slope is consistent with Lewis and Mertens (2022)
- γ_f could be larger than one and negative
- The figure in Barnichon and Mesters (2020) shows much tighter confidence sets, such as significant λ_U , because they use $\chi^2(2)$

Similar results when...

- the forcing variable is the output gap
- the sample period is 1990–2017 with the HFI proxy
- AR_a with the fixed- b critical values is used
- K_a^r and K_a are used under $\gamma_b + \gamma_f = 1$

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Fiscal policy rules (tax or spending) and the multipliers

- The fiscal policy rule

$$p_t = \psi_{gdp}^p y_t + e_t^p$$

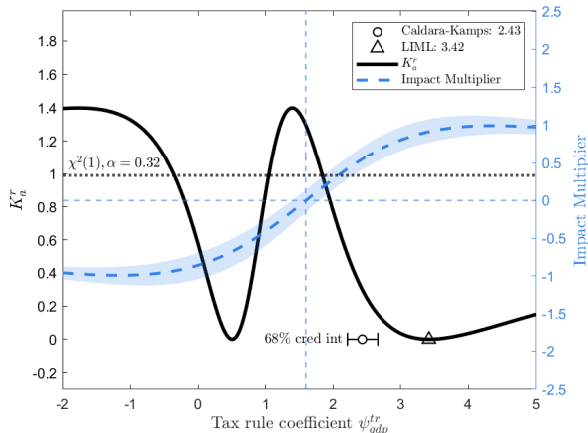
where $p \in \{tr, g\}$

- There exists a mapping between ψ_{gdp}^p and the impact multiplier given a reduced form VAR model (Caldara and Kamps 2017, CK)
- Pros: No assumption on the specification of a proxy-SVAR and identification strength
- Cons: Dynamic multiplier, different sample period between IV and data

Data

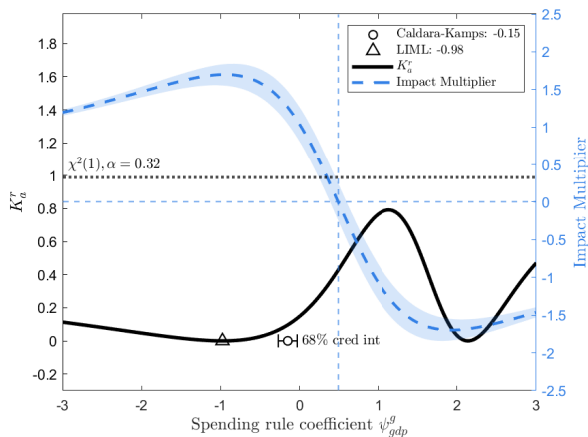
- I use the same dataset as CK
- The reduced-form VAR has 5 variables: federal tax revenue, government spending (consumption + investment), GDP, CPI, and 3-month T-Bill rate. The first three are real per capita values
- Instruments: Fernald (2012) TFP shocks
- Sample period: 1950-2006 using quarterly data
- The weak-IV test of Lewis and Mertens (2025): this proxy is not sufficiently strong ($g=3.2$, $cv = 11.4$ at the 5% significance level)

The simple tax rule and the multiplier



- The tax multiplier is significantly positive and approximately unity
- CK's credible interval is consistent with the figure, but narrower
- Short-run restriction of Blanchard and Perotti (2002) $\psi_{gdp}^{tr} = 2.08$ is not rejected (marginally), but the uncertainty is large

The simple spending rule and the multiplier



- ψ_{gdp}^g is not statistically significant, while the multiplier is, loosely speaking, likely to be above unity
- CK's credible interval is consistent with the figure, but narrower
- The restriction of Blanchard and Perotti (2002) $\psi_{gdp}^g = 0$ is not rejected, but negative ψ_{gdp}^g is equally consistent with the data

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Conclusion

Theoretically:

- ① A weak-IV robust method for overidentified macro equations
- ② HAR size control in finite samples

The recipe:

- Just-identified: Almon-parametrized AR statistic
- Overidentified: Almon-parametrized KLM statistic
- For both, use the restricted long-run variance

Empirically:

- ① The previous findings of a significant slope of the NKPC are mostly driven by size distortion
- ② A significant tax multiplier using a weak-IV robust method; however, the uncertainty remains large
- ③ Uncertainty is substantially larger than previously suggested after accounting for weak instrumental variables and size distortions

- Assumption
- Supplementary results

Assumptions I

ASSUMPTION 1. The observations $\{y_t, w_t, z_t\}$ are generated by the linear IV model

$$y_t = w_t' \delta + u_t \Leftrightarrow y = W\delta + u$$

$$w_t = z_t' \Pi + v_t \Leftrightarrow W = Z\Pi + V$$

and

$$z_t = \left(\sum_{h=0}^H \xi_{t-h}, \sum_{h=0}^H h \xi_{t-h}, \sum_{h=0}^H h^2 \xi_{t-h} \right)'$$

for $t = H + 1, \dots, n$, where δ is $m \times 1$, Π is $k \times m$ with $m \leq k = 3$, and $\eta_t = (\xi_t, u_t, v_t')$.

Theorem

Assumptions II

Further, assume that

- (i) for all t and s , we have (a) $E[\eta_t] = 0$, (b) $E[u_t \xi_s] = 0$, and (c) $E[v_t \xi_s] = 0$,
- (ii) for some $r > 2$ and finite constant Δ , we have $\sup_t \|\eta_t\|_{2r} \leq \Delta$,
- (iii) η_t is L_2 -NED (near-epoch dependence) of size $-\frac{r-1}{r-2}$ with $d_t = 1$ on V_t , where $\{V_t\}$ is an α -mixing process of size $-\frac{r}{r-2}$, Theorem

Assumptions III

- (iv) for integers $p, q \geq 0$ we have uniformly in n and H , with $H < n$, that

$$\omega_{\xi,p,n,H}^2 = \text{Var} \left[\sum_{t=H+1}^n t^p \xi_t \right] = \omega_{\xi,p}^2 (n-H)^{2p+1} + o((n-H)^{2p+1})$$

$$\Omega_{uv,q,n,H} = \text{Var} \left[\sum_{t=H+1}^n t^q (u_t, v_t')' \right] = \Omega_{uv,q} (n-H)^{2q+1} + o((n-H)^{2q+1}),$$

where $\Omega_{uv,q} = \begin{bmatrix} \omega_{u,q}^2 & \omega_{uv,q} \\ \omega_{vu,q} & \Omega_{vv,q} \end{bmatrix}$ with finite $\omega_{\xi,p}^2 > 0$ and positive definite $\Omega_{uv,q}$, and $\omega_{u,0}^2$ denotes the long-run variance of u_t and $\omega_{uv,0}$ denotes the long-run covariance of u_t and v_t . Theorem

Assumptions IV

(v) $b_n = o(n)$ and $\kappa(\cdot) \in \mathcal{K}$ where

$$\mathcal{K} = \left\{ \begin{array}{l} \kappa(\cdot) : \mathbb{R} \rightarrow [-1, 1], \kappa(0) = 1, \\ \kappa(x) = \kappa(-x) \ \forall x \in \mathbb{R}, \int_{-\infty}^{\infty} |\kappa(x)| dx < \infty, \\ \int_{-\infty}^{\infty} \left| \frac{1}{2\pi} \int_{-\infty}^{\infty} \kappa(x) e^{ivx} dx \right| dv < \infty, \\ \kappa(\cdot) \text{ is continuous at 0 and at all except for} \\ \text{a finite number of points.} \end{array} \right\}$$

Theorem

Assumptions V

Definition (Near epoch dependent (NED) stochastic processes)

A sequence of integrable random vectors $\{X_t\}$ is L_2 -NED on a stochastic sequence $\{V_t\}$ on probability space $(\Omega, \mathcal{F}, P)$ if for $m \geq 0$

$$\|X_t - E(X_t | \mathcal{F}_{t-m}^{t+m})\|_2 < d_t v_m,$$

where $\mathcal{F}_s^t = \sigma(V_s, \dots, V_t) \subset \mathcal{F}$, $t \geq s$, d_t is a sequence of non-negative constants, and $v_m \rightarrow 0$ as $m \rightarrow \infty$. We will say that the sequence is L_2 -NED of size $-s$ when $v_m = O(m^{-s-\varepsilon})$ for some $\varepsilon > 0$.

Theorem

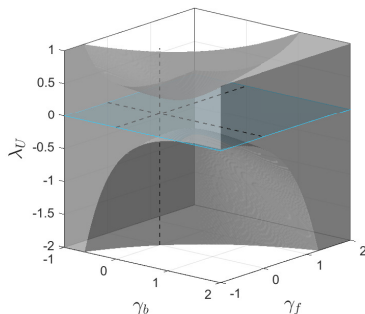
Weak-IV test: RR and HFI are weak instruments

Unemployment gap	$\gamma_b, \lambda_U, \gamma_f$		$\gamma_b + \gamma_f = 1, \lambda_U$	
	g	cv	g	cv
RR	9.3	26.0	10.9	26.0
HFI	4.6	22.9	4.8	22.9

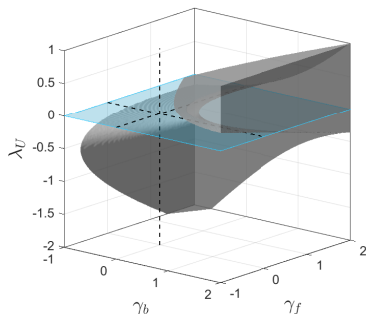
Output gap	$\gamma_b, \lambda_Y, \gamma_f$		$\gamma_b + \gamma_f = 1, \lambda_Y$	
	g	cv	g	cv
RR	5.4	25.0	5.8	25.0
HFI	8.3	21.7	8.8	21.7

Note: An instrument strength test by Lewis and Mertens (2025), testing the null hypothesis of weak instrument bias less than or equal to 10% of the worst-case benchmark. g is the test statistic, and cv is the critical value at the 5% significance level. [Back](#)

1990–2017: The HFI proxy



(a) AR_a with $\chi^2(3)$

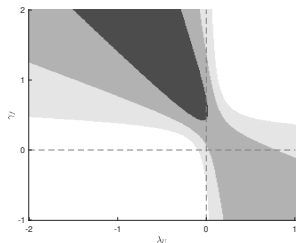


(b) AR_a^r with $\chi^2(3)$

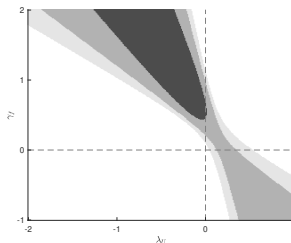
Figure: 95% confidence sets

- Similar results with the Romer-Romer proxy
- A wider region of positive λ_U . Negative γ_f, γ_b are not rejected

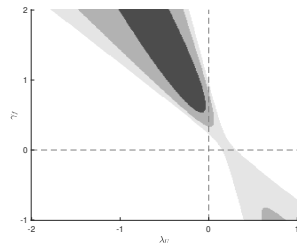
Similar results with the common restriction $\gamma_b + \gamma_f = 1$



(a) AR_a^r with $\chi^2(3)$



(b) AR_a with $\chi^2(3)$



(c) Barnichon and Mesters

- Positive λ_U and negative γ_f are more likely with AR_a^r
- The figure in Barnichon and Mesters (2020) presents much tighter confidence sets because they use $\chi^2(2)$
- Similar results with K_a^r and K_a